# Lab protocol Day 1 – gRNA cloning

Today we will clone the designed gRNA oligos into the gRNA entry vector p201. This vector contains BbsI sites for gRNA insertion by restriction digestion/ligation.

## gRNA design:

Design gRNAs using Benchling (or any other tool) and order as two short oligos. The sequences in bold are the overhangs compatible with seamless gRNA insertion into the BbsI-digested entry vector (downstream of gRNA promoter and upstream of gRNA scaffold) so **it is important that you add these overhangs to your gRNA sequences**. As we work with *S. pyogenes* Cas9, the gRNA oligos should be 20-mers.

FWD oligo gRNA: 5'-**TAGC**-forward gRNA sequence (20 nt)
REV oligo gRNA: 5'-**AAAC**-reverse complement gRNA sequence (20 nt)

Note: Adjust overhangs if you use a different promoter.

## Phosphorylation of oligos (already done):

1. Phosphorylate 5' end of each oligo separately by treating with PNK. Mix the following components

| | |
|---|---|
| Oligo (100 µM) | 0.25 µL |
| 10X T4 Ligase Buffer | 0.25 µL |
| T4 PNK | 0.25 µL |
| Sterile water | 1.75 µL |

2. Incubate at 37ºC for 1 hour

Note: It is also possible to order phosphorylated oligos (more expensive).

## Digest entry vector backbone (1h 10 min):

This step creates digested p201 entry vector with compatible overhangs for cloning the gRNA downstream of the strong constitutive J23119 promoter and upstream of the structural part of the gRNA (scaffold).

**Materials:** 1 PCR tube, p201 vector, rCutsmart buffer, BbsI-HF, sterile water
**Equipment:** vortex, mini tabletop centrifuge, thermocycler

1. Mix the following components in a PCR tube (first calculate the amount of vector needed).

| | | |
|---|---|---|
| p201 vector | 1 µg | µL |
| 10X rCutsmart | 5 µL | 5 µL |
| BbsI-HF | 1 µL | 1 µL |
| Sterile water | to 50 µL | µL |

2. Incubate at 37ºC for 1 hour.

### gRNA annealing (25 min):

This step creates double-stranded gRNA with overhangs compatible for annealing into the digested p201 entry vector.

**Materials:** 1 PCR tube per gRNA, sterile water, phosphorylated gRNA FWD and REV oligos
**Equipment:** vortex, mini tabletop centrifuge, thermocycler

1. Mix 2.5 µL FWD and REV phosphorylated oligo in a PCR tube. Add 45 µL sterile water to bring the total volume to 50 µL. Vortex and quick spin.
2. Heat 6 min at 96ºC, ramp down to 23ºC at 0.1ºC/s. Final hold at 23ºC.

Note: if you don't have a cycler with that ramp interval, it also works to just put the tube on your bench and let it cool to room temperature for 15 min.

### Clean up digested entry vector backbone (15 min):

Resulting size will be **2893 bp** for desired fragment. Clean digested product with 0.7X ratio magnetic beads. See here for more information on how bead cleanup and size selection works: http://enseqlopedia.com/2012/04/how-do-spri-beads-work/

**Materials:** 1 PCR tube, digested vector, magnetic beads (prewarmed to room temperature), 80% ethanol, sterile water
**Equipment:** magnetic stand for PCR tubes

1. Warm up beads to room temperature before starting.
2. Mix digest with 0.7X volume of magnetic beads (= 35 µL) and mix well by pipetting up and down several times (it is important that you get a homogeneously brown solution). **Be careful not to make bubbles.**
3. Incubate 5 min at room temperature, then place on magnetic stand.
4. Let beads bind to magnet for 1-2 min (or until liquid is entirely clear), then remove supernatant while leaving tubes on stand.
5. Wash **twice** with 50 µL 80% ethanol (add to beads, incubate 30s, remove EtOH, repeat)
6. Air-dry beads for 1-2 min or until ethanol has evaporated. **Take care not to dry for too long** as that will reduce yield. Ideally, the beads should no longer look 'shiny', but no cracks should be visible. **Also do not air dry too little** as residual ethanol will reduce elution efficiency.
7. Take tubes off magnet and resuspend in 25 µL of water (mix well by pipetting until homogeneously brown), incubate for 1-2 min.
8. Bind on magnet and wait until the liquid looks clear (usually 1-2 min), then transfer supernatant to a fresh tube. Transfer 1 µL less of the liquid than added for resuspension, to avoid transferring any beads.

### Determine concentration of digested and purified p201 plasmid (5 min):

**Materials**: 1 Qubit tube, dsDNA BR working solution
**Equipment:** vortex, Qubit device

1. Add 1 µL digested p201 to 199 µL 1X dsDNA BR working solution in Qubit tubes.

2. Vortex for 3 sec.
3. Incubate the tubes for 2 min at RT.
4. Insert the tubes in the Qubit ™ Fluorometer and take readings.

### Ligation (1h 10 min):

**Materials:** 1 PCR tube, annealed gRNA oligo, cleaned and digested p201, T4 ligase buffer, T4 ligase, Sterile water
**Equipment:** mini tabletop centrifuge, thermocycler

1. Mix the following components in a PCR tube (calculate first how much vector you need).

| | | |
|---|---|---|
| Digested p201 | 0.01 pmol | µL |
| Annealed gRNA oligo | 0.25 µL | 0.25 µL |
| 10X T4 ligase buffer | 1 µL | 1 µL |
| T4 DNA ligase | 0.5 µL | 0.5 µL |
| Sterile water | to 10 µL | µL |

$$\mathrm{pmol} = \frac{\mathrm{ng} \times 1{,}000}{\mathrm{bp} \times 650}$$

2. Mix well by pipetting.
3. Place tubes in thermocycler and incubate at 25ºC for 1 hour.
4. Deactivate at 65ºC for 10 minutes.

### Transformation into *E. coli* DH5α (1h 30 min):

**Materials:** ligation reaction, 1 tube chemically competent *E. coli* DH5α cells, sterile LB, 1 LB + spec plate, glass beads
**Equipment:** bucket filled with ice, heat block preheated to 42°C, tabletop centrifuge, Bunsen burner

1. Take competent cells out of -80°C and put on ice for 10 min (do while step 4 of ligation is running).
2. Carefully resuspend cells by flicking the tube.
3. Add 1 µL of ligation reaction to 15 µL of competent cells and incubate for 15 min on ice.
4. Heat shock cells for exactly 45 s in tabletop heat block @ 42°C.
5. Immediately put cells on ice for 2 min.
6. Add 1mL of sterile LB.
7. Recovery: incubate cells for 45 min at 37°C and 400 rpm in shaking incubator.
8. Spin down for 3 min at 6000 rpm.
9. Remove 900 µL of the supernatant, keeping ca. 100 µL in Eppendorf tube.
10. Resuspend pellet in this 100 µL of remaining LB.
11. Plate on LB + spectinomycin plate using glass beads – do the plating by a flame (work sterile).
12. Let plates dry by the flame and remove the beads when fully dried.
13. Incubate overnight at 37°C